Saghar Kheradmand

AI Engineer / Machine Learning Engineer / Data Scientist

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Professional Summary

M.Sc. student in Artificial Intelligence at Shiraz University with hands-on experience in machine learning, statistical pattern recognition, computer vision, image processing, evolutionary computation, reinforcement learning, backend development, and data-driven problem solving. Experienced in building reproducible Python and Jupyter workflows, implementing algorithms from scratch, evaluating models on real-world datasets, and communicating technical results clearly. Also completed a three-month internship in SEO and website design.

Education

Master of Science (M.Sc.) in Artificial Intelligence / AI Engineering — **2025–Present**

Shiraz University — *Currently studying*

- Graduate focus: machine learning, pattern recognition, intelligent systems, computer vision, neural networks, and data science.

Experience

SEO and Web Design Intern — **3-month Internship**

Practical Internship Experience

- Supported website design and content presentation tasks, with attention to usability, responsive structure, and clear page organization.
- Assisted with basic SEO activities, including on-page optimization, content structure, keyword-focused improvements, and website performance review.

Technical Skills

AI & Machine Learning	scikit-learn, from-scratch ML, neural networks, pattern recognition, PCA, LDA, QDA, Naive Bayes, classification, regression, ensemble learning, imbalanced learning
Python & Data	Python, NumPy, Pandas, SciPy, Matplotlib, Jupyter, preprocessing, feature engineering, visualization, model evaluation
Computer Vision	Edge detection, Canny, Hough transform, restoration, frequency analysis, eigenfaces, image classification, object localization
Intelligent Systems	Evolutionary computing, reinforcement learning, value-function approximation, genetic algorithms, genetic programming, search and optimization
Backend & Databases	REST APIs, Go, gRPC, SQL, PostgreSQL, MySQL, Redis, RabbitMQ, API design
Web & SEO	Website design fundamentals, responsive layout, on-page SEO, content structure, keyword optimization
Tools	Docker, Kubernetes, Linux, Git, CI/CD, unit testing, integration testing

Selected AI / Machine Learning Projects

- **Machine Learning Pipelines for Four Real-World Datasets** – Built reproducible Python/Jupyter pipelines for Airbnb, NYC 311, IBM HR, and Online Retail data, including preprocessing, missing-value treatment, feature engineering, kernel methods, and evaluation.
- **Bayesian Classification and Dimensionality Reduction** – Implemented Gaussian Naive Bayes, LDA, QDA, PCA/eigenfaces, and Fisher LDA for classification and face-recognition tasks using NumPy-based workflows.
- **From-Scratch Regression and Classification** – Developed linear and logistic regression experiments with gradient descent, evaluation metrics, confusion matrices, and visualization-driven analysis.
- **Classical Computer Vision with Hough Transforms** – Created traffic-light detection, lane-detection, and object-localization pipelines using custom edge detection, Hough voting, accumulator visualization, and post-processing.

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Selected AI / Machine Learning Projects (Continued)

- **Image Processing: Restoration and Frequency Analysis** – Built notebooks for denoising, restoration, transform-domain filtering, edge detection, frequency analysis, and visual diagnostics.
- **COVID-19 Severity Prediction with Imbalanced Learning** – Applied Hellinger Distance Decision Trees, bagging with undersampling, AdaBoost.M1, oversampling, and minority-focused evaluation metrics.
- **Ensemble Learning for Imbalanced Classification** – Compared resampling and ensemble methods on skewed datasets using confusion matrices, macro metrics, and class-sensitive model evaluation.
- **MOBGA-AOS Feature Selection** – Implemented a multi-objective genetic search method for balancing predictive performance and compact feature subsets.
- **Value-Function Approximation for Snake and Tic-Tac-Toe** – Built learning agents using handcrafted features, linear value approximation, delta-rule updates, epsilon-greedy exploration, and self-play.
- **Elman vs. Jordan Recurrent Networks** – Implemented and compared recurrent neural network variants on Tiny Shakespeare using training curves and generated text samples.
- **Neural Networks for Logical Circuits** – Modeled logical functions with neural networks and analyzed activation behavior, supervised learning, and decision boundaries.
- **Genetic Programming Maze Solver** – Designed a tree-based genetic programming system with path simulation, fitness shaping, crossover, mutation, and convergence visualization.
- **Evolution Strategies for Logistic Regression** – Used self-adaptive evolution strategies to optimize logistic-regression parameters on heart-disease data and compared results with conventional training.
- **Genetic Algorithm for Single-Machine Scheduling** – Compared crossover and mutation combinations for weighted completion-time scheduling with convergence and fitness analysis.

Selected Backend / Data Engineering Projects

- **Gym Platform Backend** – Designed a Go/Echo and PostgreSQL platform for gym entrance, locker assignment, membership tracking, reporting, JWT authentication, and role-based REST APIs.
- **Classified Advertisements REST API** – Built a Go/Echo API for ad posting, category browsing, search and filtering, pagination, authentication, permissions, and relational data modeling.
- **Payment Gateway Backend** – Developed a Go backend for customizable payment portals, user authentication, transaction management, administration, mock-bank testing, and reporting.